

Fundamental Structure & Properties of the State Transition Matrix in a Ham. Dyn. System.

- Established that the STM is a symplectic matrix.

$$\underline{\Phi} = \frac{\underline{Jx}(t)}{\underline{Jx}_0} \in \mathbb{R}^{2n \times 2n} ; \quad \bar{x}(t) = \begin{bmatrix} \vec{q} \\ \vec{p} \end{bmatrix}, \quad \underline{\dot{x}} = \underline{J} \frac{\underline{JH}}{\underline{Jx}}$$

If Φ is symplectic, then

$$\underline{\Phi} \underline{J} \underline{\Phi}^T = \underline{J}, \quad \underline{J} = \begin{bmatrix} 0 & \underline{I} \\ -\underline{I} & 0 \end{bmatrix}$$

Φ is symplectic, then
 $\Phi^T, \Phi^{-1}, \Phi^{-T}, \Phi, \Phi_2, \text{etc.} \dots$ are all symplectic.

$$\left. \begin{array}{l} |\Phi| = 1 \\ \Phi^{-1} = -\underline{J} \Phi^T \underline{J} \end{array} \right\}$$

Eigenvalues of Φ :

$|\lambda I - \Phi| = 0$, so λ exists when λ is an eigenvalue.

There are λ_i ; $i=1, 2, \dots, 2n$ ev's of Φ .

Since $\Phi \in \mathbb{R}^{2n \times 2n}$, I f λ_i is an ev, then $\overline{\lambda_i}$ is an ev.

I f λ_i is real, $\overline{\lambda_i} = \lambda_i$

I f $\lambda_i = \alpha + i\beta$, then $\overline{\lambda_i} = \lambda_{i+1} = \alpha - i\beta$

I f λ_i is an ev of Φ , then $\lambda_i^{-1} = \lambda_{i+1}$ is an ev of Φ ,

$$\lambda_i \cdot \lambda_i^{-1} = 1 \quad \checkmark$$

Assume $|\lambda I - \Phi| = 0$ show that $|\lambda^{-1} I - \Phi| = 0$ as well
 λ^{-1} exists since $|\Phi| = 1$

$$0 = |\lambda \Phi^{-1} \Phi - \Phi| = |(\lambda \Phi^{-1} - I) \Phi| = |\lambda \Phi^{-1} - I| \cdot |\overset{1}{\cancel{\Phi}}| = |\lambda \Phi^{-1} - \lambda \lambda^{-1} I|$$

$$= |\lambda (\Phi^{-1} - \lambda^{-1} I)| = \overset{\neq 0}{\cancel{|\lambda|}} |\Phi^{-1} - \lambda^{-1} I| = 0 ; \quad \Phi^{-1} = -J \Phi^T J ; \quad -J \cdot J = I$$

$$0 = |-J \Phi^T J - \lambda^{-1} (-J \cdot J)| = |J (-\Phi^T J + \lambda^{-1} J)| = \cancel{|J|} |\lambda^{-1} J - \Phi^T J| = |\lambda^{-1} I - \Phi^T| = 0$$

$$|\lambda^{-1} I - \Phi| = 0 \quad \lambda \lambda^{-1} = 1$$

If we expand the polynomial

$$\lambda^{2n} + \alpha_1 \lambda^{2n-1} + \dots + \alpha_1 \lambda + 1 = 0$$

For ev's, the most general case will

$$\lambda = \alpha + i\beta, \text{ define } \rho = \sqrt{\alpha^2 + \beta^2}, \quad \tan\theta = \frac{\beta}{\alpha},$$

$$\lambda = \rho e^{i\theta} = \rho(\cos\theta + i\sin\theta)$$

$$\text{IF } \lambda = \rho e^{i\theta} \text{ is nev, then } \bar{\lambda} = \rho e^{-i\theta}, \lambda^{-1} = \frac{1}{\rho} e^{-i\theta}, \bar{\lambda}^{-1} = \frac{1}{\rho} e^{i\theta} \Rightarrow 4 \text{ ev's}$$

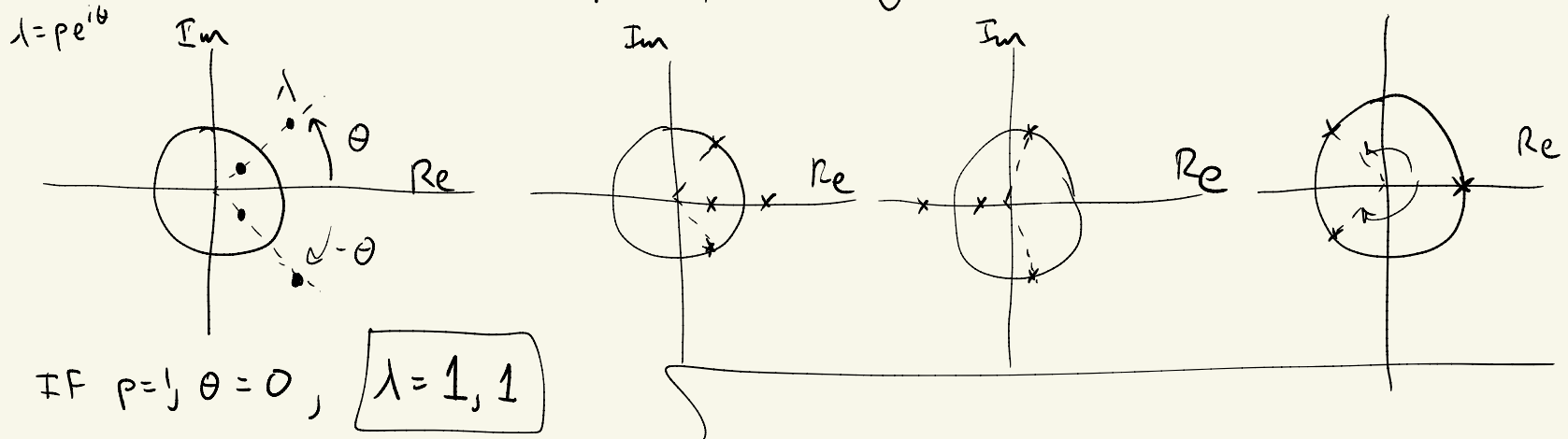
$$\text{IF } \theta = 0, \pi, e^{i\theta} = \pm 1$$

$$\lambda = \pm \rho, \lambda^{-1} = \pm \frac{1}{\rho} \Rightarrow 2 \text{ ev's}$$

$$\text{IF } \rho = 1, \lambda = e^{i\theta}, \bar{\lambda} = e^{-i\theta} = \lambda^{-1}, \bar{\lambda}^{-1} = e^{i\theta} \Rightarrow 2 \text{ ev's}$$

$$\lambda = e^{\pm i\theta} = \cos\theta \pm i\sin\theta$$

The unit circle in the complex plane organizes the eVs....



Eigenvectors ($\vec{v}$): IF $|\lambda_i I - \Phi| = 0$, then $\exists \vec{u}_i, \vec{v}_i$, right + left eigenvectors corresponding to λ_i

De find such that

$$\Phi \vec{u}_i = \lambda_i \vec{u}_i$$

right $\vec{eV}$

$$\vec{v}_i^T \Phi = \lambda_i \vec{v}_i^T$$

left $\vec{eV}$

$$\Phi^T \vec{v}_i = \lambda_i \vec{v}_i$$

Exist for
all Matrices

Note For a symplectic matrix, If $(\vec{u}_i, \lambda_i)$ are a rev pair,
 then $(J\vec{u}_i, \lambda_i^{-1})$ is a rev pair.

$$\Phi^{-1}(\Phi \vec{u}_i = \lambda_i \vec{u}_i) \quad \lambda_i^{-1} (J \vec{u}_i = \lambda_i \Phi^{-1} \vec{u}_i) \Rightarrow \boxed{\lambda_i^{-1} \vec{u}_i = \Phi^{-1} \vec{u}_i = -J \Phi^T J \vec{u}_i}$$

Mult. by J $\lambda_i^{-1} (J \vec{u}_i) = \Phi^T (J \vec{u}_i)$

In general, $\vec{v}_i^T \vec{u}_j = \delta_{ij}$ =
$$\begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$
 (assumes normalized basis)

Proof:

$$\begin{pmatrix} \vec{v}_j^T \Phi \vec{u}_i = \lambda_i \vec{v}_j^T \vec{u}_i \\ \vec{v}_j^T \Phi = \lambda_j \vec{v}_j^T \end{pmatrix} \vec{u}_i \Rightarrow \begin{pmatrix} \vec{v}_j^T \Phi \vec{u}_i = \lambda_i \vec{v}_j^T \vec{u}_i \\ \vec{v}_j^T \Phi \vec{u}_i = \lambda_j \vec{v}_j^T \vec{u}_i \end{pmatrix} \Rightarrow \boxed{(\lambda_i - \lambda_j) \vec{v}_j^T \vec{u}_i = 0}$$

If $\lambda_i \neq \lambda_j$, $\vec{v}_j^T \vec{u}_i = 0$, If $\lambda_i = \lambda_j$, no constraint

$\boxed{\vec{v}_i^T \vec{u}_i \neq 0 = 1}$

Given the ev's + ev's of Φ , the STM can be composed as:

$$\Phi = \sum_{i=1}^{2n} \lambda_i \underbrace{\vec{u}_i \vec{v}_i^T}$$

General result for all matrices
w/o repeated roots.

$$\vec{u}_i \vec{v}_i^T \in \mathbb{R}^{2n \times 2n}, \quad \underbrace{\vec{u}_i \vec{v}_i^T \text{ is singular}}$$

IF Φ is symplectic we get the general form

$$\Phi = \dots + \lambda_i \vec{u}_i (\overline{\mathcal{J} \vec{u}_{i+1}})^T + \lambda_i^{-1} \vec{u}_{i+1} (\mathcal{J} \vec{u}_i)^T + \dots \quad \text{IF } \lambda_i \text{ is real}$$

$$+ \lambda_i \vec{u}_i (\overline{\mathcal{J} \vec{u}_i})^T + \frac{\lambda_{i+1}}{\lambda_i} \vec{u}_i (\mathcal{J} \vec{u}_i)^T + \dots \quad \text{IF } \lambda_i \text{ is on the unit circle}$$

More genl if λ_i is off the unit circle + complex.